Lab 7 - Cycle index

Math 737

In [20]:
```
from IPython.display import Math, display
#------------- jupyter doesnt display sage objects properly unless I make i
def dplay(x):
    return display(Math(latex(x)))
#-------------------------------------------------------
```

Exercise 1: Use the code below to create the cyclic group of order 6. By comparing to the example from class, what does each command compute?

In [2]:
```
C6=CyclicPermutationGroup(6)
```

Creates the permutation group.

In [3]:
```
C6.conjugacy_classes()
```

Out[3]:
```
[Conjugacy class of () in Cyclic group of order 6 as a permutation group,
 Conjugacy class of (1,2,3,4,5,6) in Cyclic group of order 6 as a permutati
on group,
 Conjugacy class of (1,3,5)(2,4,6) in Cyclic group of order 6 as a permutat
ion group,
 Conjugacy class of (1,4)(2,5)(3,6) in Cyclic group of order 6 as a permuta
tion group,
 Conjugacy class of (1,5,3)(2,6,4) in Cyclic group of order 6 as a permutat
ion group,
 Conjugacy class of (1,6,5,4,3,2) in Cyclic group of order 6 as a permutati
on group]
```

Returns the conjugacy classes of the permutation group.

In [23]:
```
a=C6.cycle_index()
dplay(a)
```

$\displaystyle \frac{1}{6} p_{1,1,1,1,1,1} + \frac{1}{6} p_{2,2,2} + \frac{1}{3} p_{3,3} + \frac{1}{3} p_{6}$

Returns the cycle index polynomial. (This is the sum of the table we created using Burnside's Lemma.)

In [24]:
```
dplay(C6.profile_series())
```

$\displaystyle z^{6} + z^{5} + 3 z^{4} + 4 z^{3} + 3 z^{2} + z + 1$

Returns the generating function for the number of orbits.

Exercise 2: Use similar commands on the dihedral group to compute its conjugacy classes, cycle index, and profile series. How many 6 bead bracelets with 2 colors are there if bracelets are considered to be the same under rotation or reflection? How many of these have 3 beads of each color?

In [6]:
```
D6 = DihedralGroup(6); D6
```

Out[6]:
```
Dihedral group of order 12 as a permutation group
```

In [7]:
```
D6.conjugacy_classes()
```

Out[7]:
```
[Conjugacy class of () in Dihedral group of order 12 as a permutation grou
p,
 Conjugacy class of (2,6)(3,5) in Dihedral group of order 12 as a permutati
on group,
 Conjugacy class of (1,2)(3,6)(4,5) in Dihedral group of order 12 as a perm
utation group,
 Conjugacy class of (1,2,3,4,5,6) in Dihedral group of order 12 as a permut
ation group,
 Conjugacy class of (1,3,5)(2,4,6) in Dihedral group of order 12 as a permu
tation group,
 Conjugacy class of (1,4)(2,5)(3,6) in Dihedral group of order 12 as a perm
utation group]
```

In [25]:
```
b=D6.cycle_index()
dplay(b)
```

$\displaystyle \frac{1}{12} p_{1,1,1,1,1,1} + \frac{1}{4} p_{2,2,1,1} + \frac{1}{3} p_{2,2,2} + \frac{1}{6} p_{3,3} + \frac{1}{6} p_{6}$

In [26]:
```
dplay(D6.profile_series())
```

$\displaystyle z^{6} + z^{5} + 3 z^{4} + 3 z^{3} + 3 z^{2} + z + 1$

In [22]:
```
dplay(D6.cycle_index().expand(2))
#Answer is the sum of coefficients: 0 + 1 + 3 + 3 + 3 + 1 + 0 = 11
#There are three distinct necklaces that have 3 beads of each color (coeffic

#For my own reference add .subs(x1=1).subs(x0=2) [cannot be used in conjunct
```

$\displaystyle x_{0}^{6} + x_{0}^{5} x_{1} + 3 x_{0}^{4} x_{1}^{2} + 3 x_{0}^{3} x_{1}^{3} + 3 x_{0}^{2} x_{1}^{4} + x_{0} x_{1}^{5} + x_{1}^{6}$

Exercise 3: Consider 2-colorings of the faces of the cube, as in the Sagan textbook Section 6.2. Type in the permutation group generators given in the table in the middle of p. 196 to generate a permutation group (there is an example below of how to do this). Use the commands from the prior exercise to compute the number of orbits under this group action with all possible combinations of the number of faces of each color. Check that your answers match the computations in the book.

In [11]:
```
p = Permutation([(1,2),(3,4)]); p
```

Out[11]:
```
[2, 1, 4, 3]
```

In [29]:
```
pgroup = PermutationGroup([p]); pgroup
```

Out[29]:
```
Permutation Group with generators [(1,2)(3,4)]
```

p.196 Permutation Group Generators:

In [32]:
```
e=Permutation([(1),(2),(3),(4),(5),(6)])
g1 = Permutation([(2, 3, 4, 5)])
g2 = Permutation([(2, 4), (3, 5)])
g3 = Permutation([(1, 4), (2, 6), (3, 5)])
g4= Permutation([(1,2,5), (3,6,4)])
G = PermutationGroup([e,g1,g2,g3,g4])
display(G)
```

```
Permutation Group with generators [(), (2,3,4,5), (2,4)(3,5), (1,2,5)(3,6,
4), (1,4)(2,6)(3,5)]
```

In [33]:
```
G.conjugacy_classes()
```

Out[33]:
```
[Conjugacy class of () in Permutation Group with generators [(), (2,3,4,5),
(2,4)(3,5), (1,2,5)(3,6,4), (1,4)(2,6)(3,5)],
 Conjugacy class of (2,3,4,5) in Permutation Group with generators [(), (2,
3,4,5), (2,4)(3,5), (1,2,5)(3,6,4), (1,4)(2,6)(3,5)],
 Conjugacy class of (2,4)(3,5) in Permutation Group with generators [(),
(2,3,4,5), (2,4)(3,5), (1,2,5)(3,6,4), (1,4)(2,6)(3,5)],
 Conjugacy class of (1,2)(3,5)(4,6) in Permutation Group with generators
[(), (2,3,4,5), (2,4)(3,5), (1,2,5)(3,6,4), (1,4)(2,6)(3,5)],
 Conjugacy class of (1,2,3)(4,5,6) in Permutation Group with generators
[(), (2,3,4,5), (2,4)(3,5), (1,2,5)(3,6,4), (1,4)(2,6)(3,5)]]
```

In [35]:
```
dplay(G.cycle_index())
dplay(G.profile_series())
```

$\displaystyle \frac{1}{24} p_{1,1,1,1,1,1} + \frac{1}{8} p_{2,2,1,1} + \frac{1}{4} p_{2,2,2} + \frac{1}{3} p_{3,3} + \frac{1}{4} p_{4,1,1}$

$\displaystyle z^{6} + z^{5} + 2 z^{4} + 2 z^{3} + 2 z^{2} + z + 1$

Plugging in 1 to the profile series will give you the number of orbits of all possible n-element subsets in this action, which which is just the total number of orbits since the set being acted on here is finite. The result is confirmed on the next page.

In [37]:
```
dplay(G.profile_series().subs(z=1))
```

$\displaystyle 10$

Exercise 4: Find the number of elements in the group you made in Exercise 3. Guess what permutation group this group is isomorphic to. Check your guess by using the command .isomorphism_to()

In [38]:
```
G.cardinality()
```

Out[38]: 24

This group is not abelian since $(12)(35)(46)\circ (24)(35)= (1264)\neq (1462)= (24)(35)\circ(12)(35)(46)$. So it's either D_{12} or S_{4} up to isomorphism. But no element has order 12 since order of permutations in cycle product forms anyways is the lcm of its cycle lengths none of which multiply to 12 here. So it's S_{4}. This confirmed below.

In [39]:
```
G.is_isomorphic(SymmetricGroup(4))
```

Out[39]: True